EchoForge: A Strict-Open Synthetic Benchmark for Low-Altitude UAS Radar and Multimodal Sensing

EchoForge Project

***Abstract*—EchoForge is a strict-open synthetic sensing benchmark for low-altitude uncrewed aerial system (UAS) radar and multimodal detection research. Strict-open means that assumptions, code, configuration, detector-view schemas, and paper evidence are reviewable from public or generated sources, while generated arrays and measured raw files remain outside Git. The paper-facing positive class is a fixed-wing pusher-prop public proxy with explicit uncertainty, not a measured-platform claim. EchoForge couples a radar signal chain, acoustic and passive-RF cue definitions, group-locked scenario splitting, detector-view feature denylisting, false-alarm-family analysis, and holdout-isolated evaluation into a reproducible evidence package. The v2 single-seed run contains 10,000 scenario groups, 50 positive groups, and 30,000 phase records. On the 4,500-record blind holdout, with 24 positive records from eight positive scenario groups, the locked candidate reaches point Recall@$\leq$1%FPR of 0.833 and LCB95 Recall@$\leq$1%FPR of 0.699; the prior fusion baseline reaches 0.292 and 0.083, respectively. The locked candidate improves AP and low-FPR recall while its ROC AUC is lower than the prior fusion baseline. These results are synthetic public-proxy benchmark evidence under declared assumptions, not measured truth or operational performance claims.**

***Index Terms*—Synthetic radar, UAS detection, public proxy, micro-Doppler, calibration, reproducibility, compare-only measured anchor.**

I. Claim Boundary and Contributions

EchoForge is intentionally narrower than a measured radar collection. It publishes public-proxy signatures and detector-facing products with explicit uncertainty and validation tiers, while keeping generated arrays, solver outputs, and measured raw files out of Git. The radar and detection framing follows standard signal-processing and CFAR practice rather than a sensor-surrogate claim [1]–[3]. The paper uses repository-compatible legacy identifiers only where required for tooling; the paper-facing positive class is named *fixed-wing pusher-prop public proxy*. No measured-platform truth, proprietary-equivalent behavior, or classified fidelity is claimed for any object, platform, or sensor.

The upgrade in this paper is substantive rather than cosmetic. We move from a governance-heavy benchmark report to a radar-first paper with:

- a signal-chain model card with nominal bands, bandwidths, PRF/CPI, pulse counts, range bins, and resolution assumptions;
- scenario-balance and leakage diagnostics that cover split $\times$ site/range/aspect/noise/hard-negative family counts, marginal imbalance scores, a stratum-only baseline, a metadata-only baseline, a label-shuffle sanity test, and a nearest-neighbor train/holdout scan;
- holdout metrics with counts, fixed-FPR recall, calibration, Brier score, ECE, and group-block bootstrap confidence intervals;
- engineered-intelligence transparency for the advanced fusion candidate, with reader-facing modality and ablation summaries and locked evidence artifacts in the appendix bundle;
- public-source detector archetype cards and a regional bird/RC hard-negative taxonomy for low-FPR false-alarm analysis;
- a compare-only KTH anchor overlay for hard-negative realism and observable-shape checks.

Benchmark positioning is deliberately compare-only. KTH provides a public 77 GHz FMCW measured anchor for drone/bird/human observable-shape checks; Open Radar and RAD-DAR/RDRD provide public radar and range-Doppler reference points; and public UAS micro-Doppler studies bound the observable families used for small aerial objects [4]–[10]. EchoForge does not replace those measured collections. It provides a strict-open synthetic lane for controlled hard negatives, leakage checks, and low-FPR detector comparison when measured positive-class truth is not available.

Figure 1 summarizes the signal chain and evidence boundary.

II. Radar and Multimodal Generative Model

EchoForge uses a clean-room signal-chain proxy rather than a proprietary sensor surrogate. The received-power proxy is written as

$$P_r \propto \frac{P_t G_t G_r \lambda^2 \sigma(\theta, f, \phi)}{(4\pi)^3 R^4 L}, \tag{1}$$

where the radar cross section σ is an aspect- and frequency-dependent public-proxy quantity, R is slant range, and L is an aggregate loss term. Complex IQ is generated as a phase-stable target term plus clutter, interference, and receiver impairment terms,

$$x_{m,n} = a_{m,n} e^{j(2\pi f_d m T_r + \varphi_n)} + c_{m,n} + \epsilon_{m,n}, \tag{2}$$

with m indexing pulses and n indexing range bins. The benchmark keeps these values synthetic: the absolute scale is a proxy, and the public-proxy object family is defined by priors rather than measured signatures, as in conventional micro-Doppler and small-UAS radar modeling [8]–[13].

The range-Doppler proxy uses FMCW/chirp wording for the synthetic range product and pulse/CPI wording for Doppler indexing. The derived quantities are

$$\Delta R = \frac{c}{2B}, \qquad \Delta f_D = \frac{1}{T_{\mathrm{CPI}}}, \qquad \Delta v = \frac{\lambda}{2T_{\mathrm{CPI}}}, \tag{3}$$

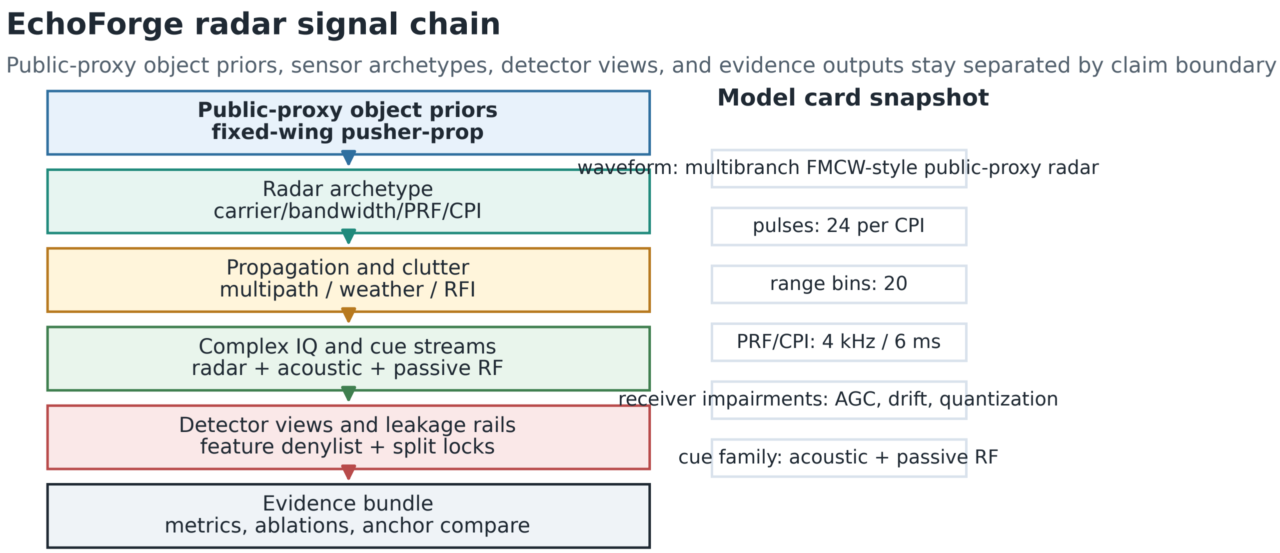


Fig. 1. Radar signal chain and claim boundary. Public-proxy object priors, radar archetype assumptions, clutter and artifact priors, detector views, and evidence outputs are separated so the benchmark can be audited without conflating synthetic products with measured truth.

with approximate unambiguous velocity $\lambda f_{\mathrm{PRF}}/4$ for the simplified branch cards. Target priors include an aspect-dependent RCS envelope, Swerling-like fluctuation proxy, 150–217 Hz pusher-prop micro-Doppler proxy, and hard-negative bird/RC cadence bands. Clutter/noise/RFI priors cover Weibull and K-like clutter, sea/terrain/weather stress, AGC compression, clock drift, quantization, dropped CPI, PRF ambiguity, Doppler folding, calibration offset, multipath masking, and RFI burst regimes. Table I is the compact model card used by the evidence bundle. The values are public-proxy assumptions, not claims about any operational sensor.

The receiver impairment set includes AGC compression, clock drift, quantization, dropped CPIs, PRF ambiguity, Doppler folding, calibration offset, and multipath masking. Acoustic cues are modeled as node-wise spectral cadence and inter-node agreement; passive-RF cues are modeled as no-signal, RFI burst, clock-offset, and provenance-quality geometry. These definitions are public-proxy abstractions, not measured claims, and are aligned with the literature on measured-anchor comparison rather than truth transfer [4]–[7].

III. Public-Source Detector Archetypes

EchoForge detector branches are public-source archetype envelopes. The public examples are used to bound branch roles and observable families, not to reproduce vendor processing. The X/Ku C-UAS branch is anchored by public descriptions of low-slow-small radar behavior such as Blighter A400 and KuRFS [14], [15]. The tactical S-band branch follows public MHR/nMHR-style hemispheric surveillance roles [16]. The GBAD cueing branch follows public wide-area 3D/4D radar roles such as Giraffe 1X and Sentinel summaries [17], [18]. In all cases, "on par" means plausible public branch role and observable family, not sensitivity, ECCM, proprietary clutter rejection, exact Pd/Pfa, or military performance.

IV. Scenario Design and Detector Views

The main run consists of 10,000 scenario groups, 50 positive groups, and 30,000 phase records generated from a single seed. Each group expands into three locked windows: initial take-up, climb transition, and cruise altitude. Split assignment occurs at the scenario-group level so all phase records from one group share the same split role and fold. In the v2 split, the default holdout contains 1,500 groups with 8 positive groups, leaving 42 positive groups for train/CV; every train/CV fold receives positives, and the default benchmark remains group-random holdout. The paper evidence also supports a targeted holdout policy for diagnostic out-of-distribution slices.

Figure 2 shows the group-locked split and the leakage diagnostics that accompany it. The paper evidence records label-split counts, site/range/aspect/noise counts, hard-negative family counts, marginal imbalance scores, a stratum-only baseline, a metadata-only baseline, a label-shuffle sanity test, a forbidden-feature canary test, and a nearest-neighbor train/holdout similarity scan.

The detector-facing views remain intentionally narrower than the generator state. The high-resolution X/Ku, tactical S-band, and GBAD branches expose radar-specific numeric summaries; the acoustic branch exposes cue statistics; the passive-RF branch exposes no-signal and provenance-quality geometry; and the fusion branch exposes calibrated late-fusion scores. The evidence bundle now emits a detector-view schema table listing allowed feature families and audit-only fields. Restricting the model view prevents labels, split keys, group

TABLE I
RADAR MODEL CARD AND DERIVED RESOLUTION ASSUMPTIONS

Branch	f_0 GHz	BW MHz	PRF/CRF Hz	CPI ms	Chirps	Span m	ΔR m	Δv m/s
X/Ku C-UAS	10.0	600	4000/4000	6.0	24	5.0	0.250	2.498
S-band AESA/MHR	3.1	180	4000/4000	6.0	24	16.7	0.833	8.059
GBAD 3D/4D	10.3	300	4000/4000	6.0	24	10.0	0.500	2.426

TABLE II
PUBLIC DETECTOR ARCHETYPE CARDS

Branch	Public role envelope	Detector-visible products	Excluded truth fields
X/Ku C-UAS	Low-slow-small search, high-resolution range-Doppler, bird/RC discrimination	Range-Doppler concentration, micro-Doppler spread, clutter stress	Classified sensitivity, proprietary clutter maps, exact Pd/Pfa
Tactical S-band AESA/MHR	Hemispheric surveillance, track-while-scan, medium tactical cueing	Coarser velocity bins, track confidence, multipath stress	Vendor processing, ECCM, deployment geometry
GBAD 3D/4D	Wide-area cueing, revisit delay, radar-horizon stress	Cue confidence, stale-track state, horizon masking	Engagement-quality behavior, operator workflows
Acoustic network	Spectral cadence and cross-node agreement	Propulsion cadence, bearing/time consistency, weather noise	Exact microphone layout, field localization performance
Passive RF	No-signal behavior, RFI bursts, provenance quality	Missingness, sparse RF cues, RFI/provenance flags	Emitter libraries, operator identity, payload commands
Fusion C2	Source confidence, cross-sensor confirmation, false-track accounting	Late-fusion score, source quality, stale-track penalty	Operational C2 logic or deployment topology

TABLE III
VALIDATION TIERS AND CURRENT STATUS

Tier	Status	Boundary
Synthetic corpus generation	pass	Scenario groups, detector views, and raw products are generated from a reproducible seed
Detector split isolation	pass	Holdout rows are excluded from calibration, thresholding, and candidate selection
Compare-only measured anchor	pass	KTH is used only for distribution-shape checks and hard-negative realism
Generative-origin self-classification	pass	Manifest-based LOC audit reports an auditable self-classification, not authorship proof
Named-platform truth claim	prohibited	No measured-object equivalence or operational performance claim is made

TABLE IV
REGIONAL HARD-NEGATIVE FAMILIES

Family	Proxy band	Main confusion mechanism
Shorebird/wader	5–14 Hz wingbeat	Small RCS, dense migration, low coastal returns
Gull/tern	2.5–8 Hz wingbeat	Moderate body returns and coastal flocking
Raptor/falcon	1.5–6 Hz wingbeat	Fast individual tracks and glide/flap transitions
Flamingo/large bird	1.2–4.5 Hz wingbeat	Larger body return, slower wingbeat, wetland corridors
Seabird/cormorant	2–7 Hz wingbeat	Sea-clutter interaction and low marine tracks
RC fixed-wing	40–220 Hz prop cadence	Prop cadence, speed, and aspect overlap with positives

identifiers, and hidden generator metadata from becoming shortcuts.

The hard-negative roster is regionalized for Gulf/coastal site archetypes. Bird stressors are split into single bird, flock aggregate, shorebird/wader, gull/tern, raptor/falcon, flamingo or other large bird, seabird/cormorant, and seasonal migratory density families. The taxonomy uses public biodiversity context for regional prevalence and published drone/bird micro-Doppler behavior for wingbeat and spectral-overlap assumptions [8], [9], [19]. RC fixed-wing aircraft are modeled as a separate benign hard-negative family: hobby-class airframes, prop cadence overlap, short-to-medium range, variable maneuvering, and speed/aspect overlap with the positive public proxy. They are not target proxies.

V. EVALUATION PROTOCOL AND CALIBRATION

Let s_i be the selected score for record i. For the locked advanced candidate, the fused score is a nonnegative weighted

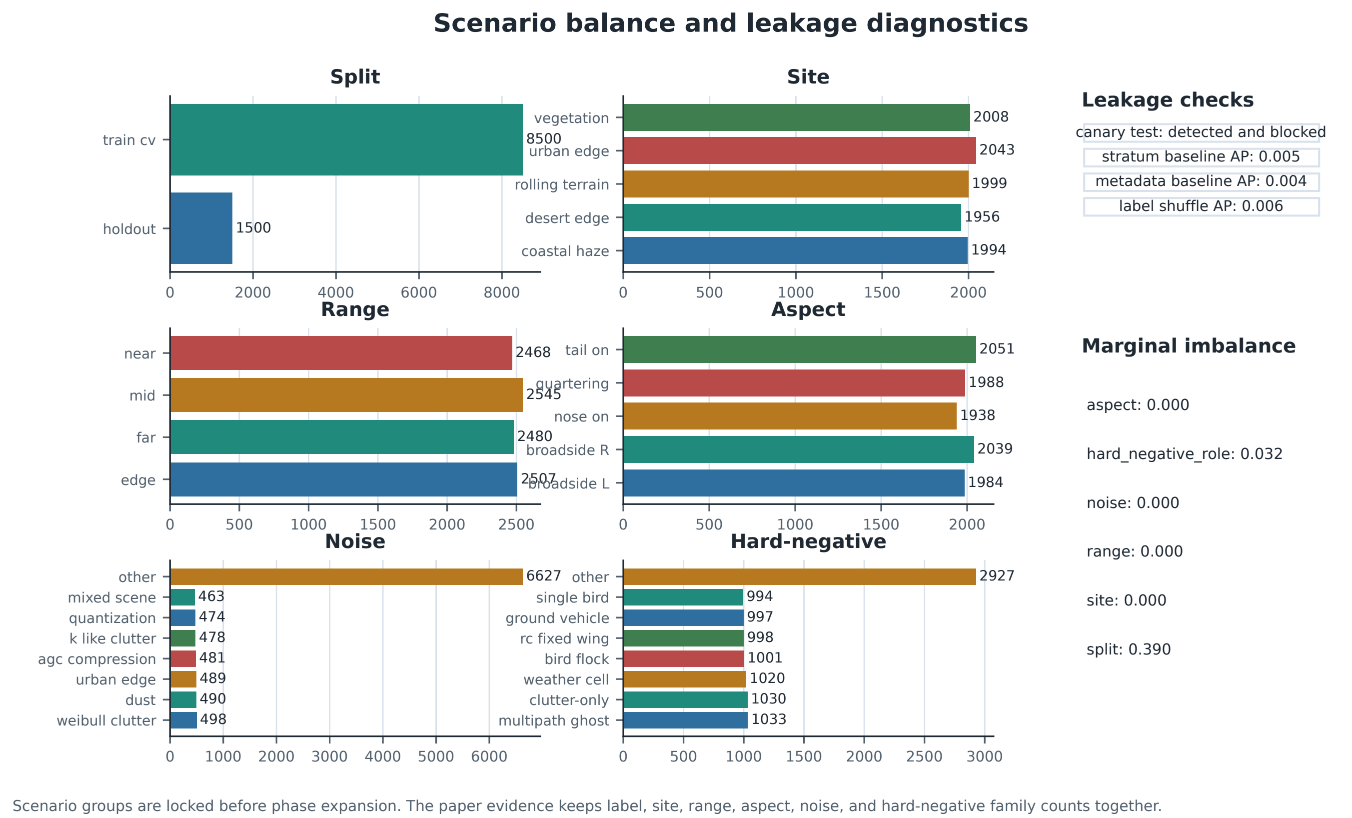


Fig. 2. Scenario balance and leakage diagnostics. The evidence bundle keeps split, label, site, range, aspect, noise, and hard-negative family counts together with canary and sanity checks. Audit-only split/group headers are reported as detected and blocked by the model-facing schema rather than as detector features.

sum over component scores,

$$z_i = \sum_{j=1}^{k} w_j s_{ij}, \qquad w_j \geq 0, \qquad \sum_{j=1}^{k} w_j = 1. \tag{4}$$

The paper-facing calibration step is a geodesic-odds transform implemented as a monotone log-odds map,

$$\hat{p}_i = \sigma\left(a \log \frac{z_i + \epsilon}{1 - z_i + \epsilon} + b\right), \tag{5}$$

where σ is the logistic function and ϵ prevents boundary singularities. We use train/CV rows only to fit thresholds and calibrators, and we score holdout rows once after the selection lock is written.

The main metrics are ROC AUC, average precision, fixed-FPR recall, accuracy, precision, recall, false-positive rate, F1, Brier score, and expected calibration error (ECE). Group-block bootstrap confidence intervals are computed by resampling scenario groups, not individual phase rows, so that uncertainty reflects the locked group structure rather than an artificially inflated sample size [20]–[23]. The paper evidence bundle also records PR and ROC curve traces and calibration bins, which matches standard classifier-evaluation and calibration practice [24]–[31].

A. Primary KPI

The headline KPI is the lower 95% group-block bootstrap bound of recall at FPR $\leq$ 1%, abbreviated *LCB95 Recall@≤1%FPR*. That choice is deliberate: the benchmark is rare-positive, false-alarm-constrained, and group-correlated, so the raw point estimate of recall is useful but not sufficient as the primary decision metric. AP, ROC AUC, raw recall, precision, F1, calibration, and false-alarm breakdowns remain diagnostic guardrails rather than the headline KPI.

Figure 3 shows the primary KPI first and keeps AP, ROC/PR curves, and calibration as supporting diagnostics. Table VI summarizes the prior fusion baseline and the locked advanced candidate on the blind holdout split. In the v2 run, the holdout split contains 4,500 records, of which only 24 records from 8 positive scenario groups are positive. The selected candidate improves AP, selected-threshold precision, false-positive control, and swept Recall@≤1%FPR over the prior layered fusion baseline. Its ROC AUC is lower than the prior fusion baseline, so the result should be read as better low-FPR operating behavior for this benchmark rather than universal rank dominance. Because the positive holdout has $n = 8$ groups, phase and family conclusions are diagnostic.

The evidence bundle also records the canary forbidden-feature test, the label-shuffle sanity test, the stratum-only baseline, the metadata-only baseline, and the nearest-neighbor train/holdout scan. Those tests are not leaderboard scores; they are leakage and realism diagnostics that make the benchmark harder to overfit.

Figure 4 highlights that the initial take-up window remains the hardest phase, while the cruise window is the most separa-

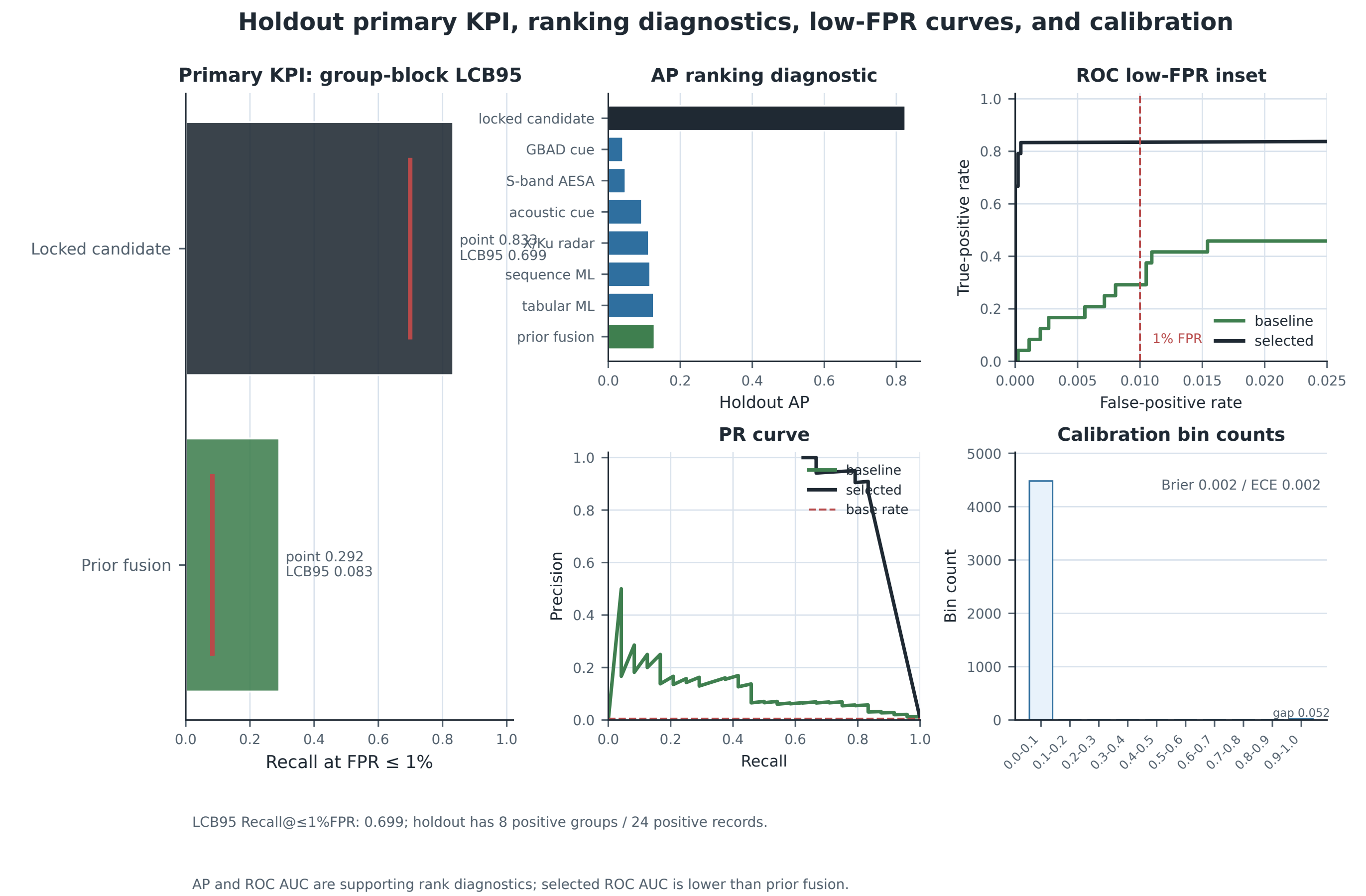


Fig. 3. Holdout primary KPI, ranking diagnostics, low-FPR ROC/PR behavior, and calibration. Recall@$\leq$1%FPR and its LCB95 are the headline quantities; AP, ROC AUC, PR shape, and calibration bins are supporting diagnostics.

ble. The cross-method false-positive burden shows whether the strongest methods overreact to birds, RC fixed-wing aircraft, weather, clutter-only counterfactuals, or other hard-negative families. The point of the breakdown is not to optimize evasion; it is to expose robustness failure modes.

VI. Engineered Intelligence

The advanced lane is reported as engineered intelligence rather than as an opaque model handle. The reader-facing story is train/CV-only candidate discovery, evolved feature families, sparse calibrated fusion, and a single blind holdout score pass after the lock. The evidence bundle keeps the raw component handles and search traces in appendix artifacts, while the paper foregrounds the stage summaries that matter for review.

Figure 5 replaces an opaque winner label with a transparent review surface. The component weights are reader-facing summaries of the locked sparse fusion, while the ablation/control deltas use the same holdout labels and metric functions as Table VI. Values copied directly from `selected_component_ablations.csv` are treated as ablation-internal diagnostics unless they share that metric basis.

VII. Raw Samples and Compare-Only Anchor

The raw sample figures are visual diagnostics, not detector features. Figure 6 exposes the model card and signal-chain assumptions, while Figure 7 shows range-Doppler proxy views for a positive public-proxy group and a hard-negative group across the three phase windows.

The compare-only measured anchor is KTH, a 77 GHz FMCW drone/bird/human collection that is useful for distribution-shape checks but not for positive-class truth claims [4]. Figure 8 shows normalized KTH anchor medians and q10/q90 intervals as measured-anchor z-scores, avoiding mixed-unit bars. The purpose is to check whether the synthetic hard-negative observables land in a sensible micro-Doppler and range-style neighborhood, not to promote the public-proxy fixed-wing class into a measured-platform claim; public anchors such as Open Radar, RAD-DAR, and related datasets are used in the same compare-only spirit [5]–[7].

VIII. Compact Diagnostics Snapshot

The evidence bundle also contains compact diagnostic summaries that are easier to audit in table form than in figure captions. Table XI shows the leakage and realism checks, Table XII shows the selected-threshold confusion matrix, Table XIV shows the selected candidate by phase, and Table XIII

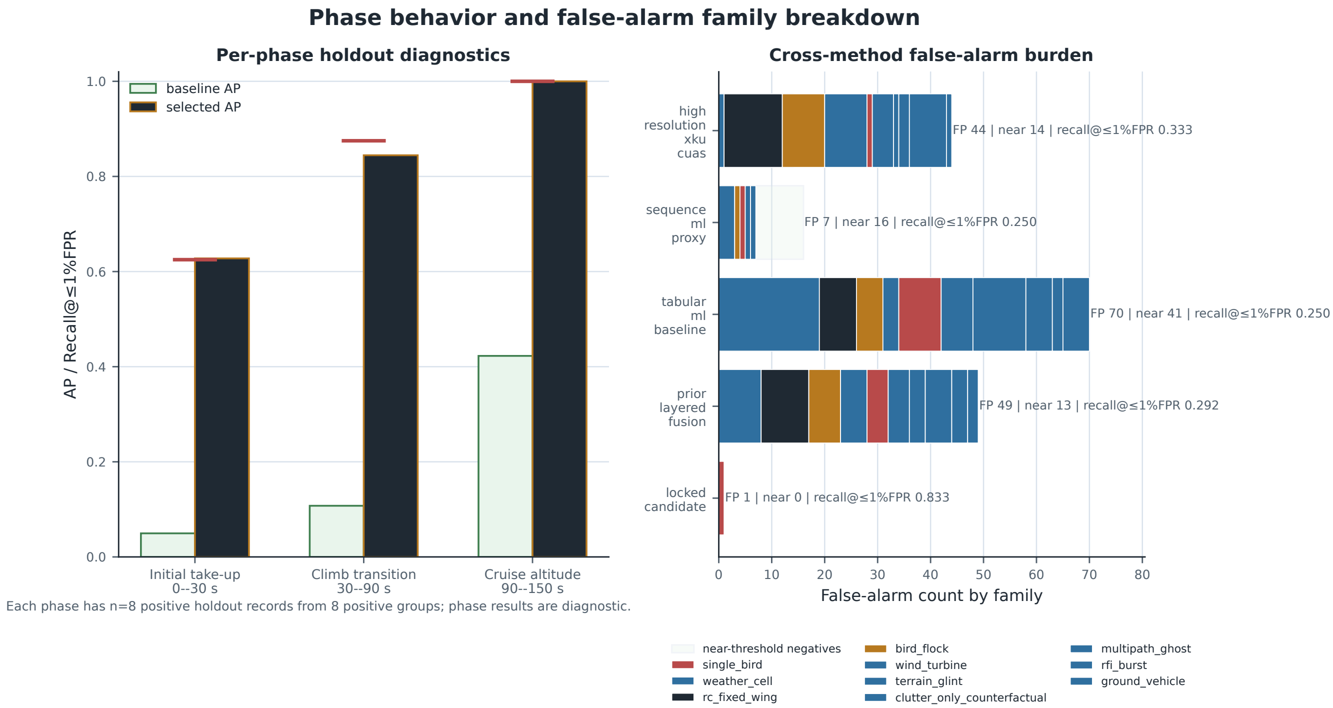


Fig. 4. Per-phase behavior and cross-method false-positive burden. The left panel compares phase-wise AP and fixed-FPR recall for the baseline and locked candidate; the right panel sorts false alarms by family for the strongest methods together with near-threshold context. Each phase has only eight positive holdout records from eight positive groups, so phase conclusions are diagnostic.

TABLE V
PRIMARY KPI SELECTION FOR THE LOCKED HOLDOUT EVALUATION

Metric	Role	Why it is or is not the headline
LCB95 Recall@$\leq$1%FPR	Primary	Captures the low-FPR operating behavior under group-block uncertainty, which matches the benchmark decision problem.
AP	Secondary	Useful for ranking, but it can obscure low-FPR behavior under extreme class imbalance.
ROC AUC	Secondary	Stable as a rank diagnostic, but insensitive to the low-FPR operating regime that matters here.
Brier / ECE	Secondary	Important for calibration and reliability, but not a direct operating-point safety metric.
False-alarm families	Secondary	Essential for robustness interpretation, but not a scalar KPI by itself.

summarizes the top-method false-positive burden. Table XV remains a cautious locked-candidate family diagnostic.

A. *How to Read the Diagnostics*

The raw audit-header check is deliberately run before the model-facing denylist is applied. It confirms that split and group metadata are visible to audit code and blocked before modeling, which is the behavior needed to prove the guard rather than assume it.

The label-shuffle test and the weak metadata baselines provide a separate sanity check. When the labels are permuted, average precision falls from 0.825 to 0.006 and ROC AUC falls from 0.917 to 0.497, which is the collapse expected if the model is responding to detector evidence rather than index structure. The stratum-only and metadata-only baselines are not meant to compete with the detector; they are meant to show that site, range, aspect, and noise metadata alone do not explain the observed ranking performance.

The phase table and the false-alarm tables fill in the operational interpretation. The initial take-up window is still the hardest phase, which is the right failure mode for a radar-oriented benchmark because the target is still transitioning and the observable is least settled. The false alarms are not spread uniformly across every negative family; instead, they concentrate in `single_bird` for the locked candidate and in the bird and RC families for the stronger baseline methods. That is a helpful robustness diagnostic because it identifies where the benchmark can still be stressed without needing any proprietary or operational sensor claims.

Finally, the engineered-intelligence stage table, component weights, and comparable view-control rows are audit trails, not claims about a universal ranking of features. They show what the locked meta-fusion candidate uses and keep ablation-

TABLE VI
HOLDOUT SUMMARY SEPARATING RANKING, SELECTED-THRESHOLD COUNTS, AND SWEPT FIXED-FPR METRICS

Method	AP	ROC AUC	F1	ECE	Threshold source	TP/FP/FN	Sel. FPR	Recall@$\leq$1%FPR	LCB95
layered_fusion_c2	0.128	0.938	0.241	0.016	train/CV max-F1	10/49/14	0.010947	0.292	0.083
locked candidate	0.825	0.917	0.837	0.002	selected lock	18/1/6	0.000223	0.833	0.699

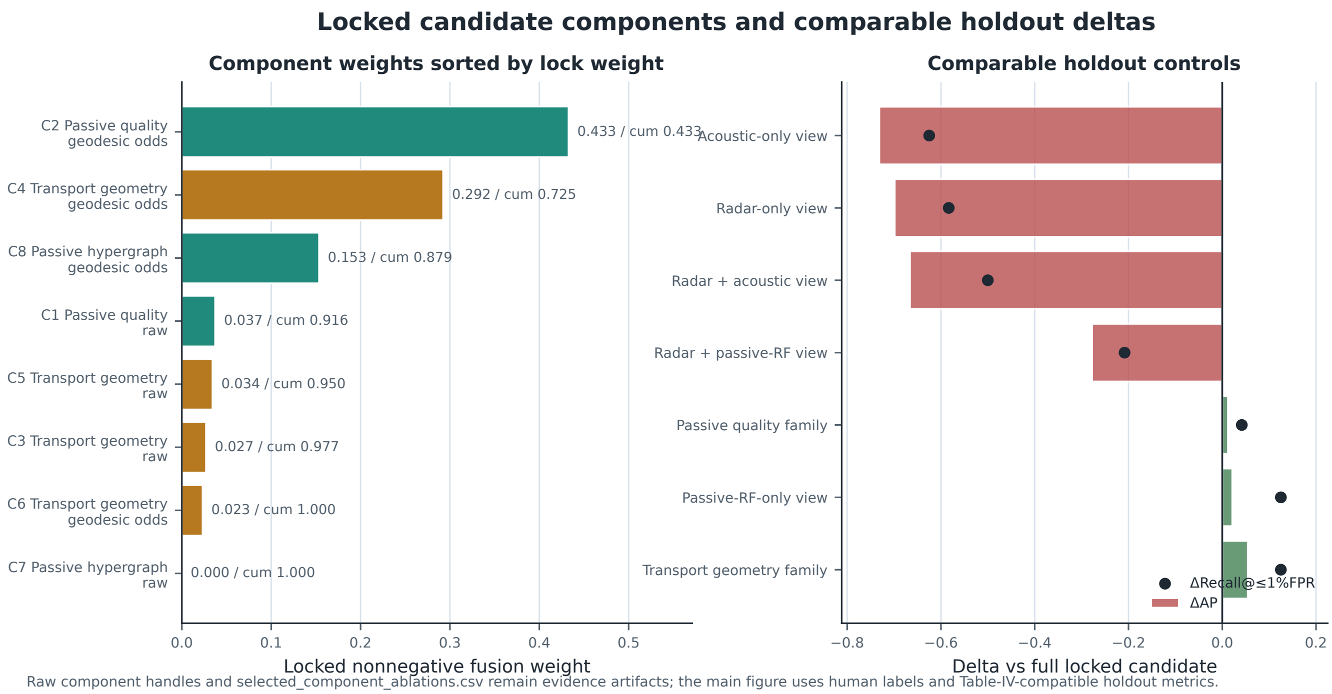


Fig. 5. Locked-candidate component weights and comparable holdout controls. The left panel sorts human-readable component weights; the right panel reports AP and fixed-FPR recall deltas computed with the same holdout metric definitions as Table VI. Raw component handles and ablation-internal diagnostics remain in evidence artifacts.

TABLE VII
ENGINEERED INTELLIGENCE TRANSPARENCY SUMMARY

Stage	Reader-facing summary	Status
Agentic decomposition	Multiple train/CV branches are explored before any holdout scoring occurs.	pass
Train/CV-only candidate discovery	Ranking, calibration, and thresholding stay on train/CV rows.	pass
Evolved feature families	Sparse radar, passive-RF, and transport-style families contribute to the lock.	pass
Sparse calibrated fusion	A small nonnegative weight vector is calibrated with a monotone odds map.	pass
Holdout-only final scoring	The blind holdout is scored once after the lock, with no further tuning.	pass

internal scores separate from Table-IV-compatible holdout metrics. That makes the locked-candidate result interpretable in the same way the compare-only KTH overlay is interpretable: as evidence about the benchmark and detector behavior, not as a platform truth claim.

IX. EVIDENCE LEDGER

The paper evidence bundle is intentionally split into a small number of human-readable artifacts so that a reviewer can reconstruct the claim boundary without chasing generator internals. The radar model card and detail rows define the public-proxy carrier bands, bandwidths, CPI, PRF/CRF, chirp count, range span, range/Doppler/velocity resolutions, unambiguous-velocity proxy, clutter/noise/RFI priors, RCS prior, and cue definitions. Those are not operational sensor facts; they are explicit synthetic assumptions that make the signal chain legible and make the detector outputs comparable across runs.

The scenario balance files describe the split structure at the scenario-group level, including split, site, range, aspect, weather/noise, hard-negative family, phase, and positive-only balances. That choice matters because a row-level split would allow the same physical group to leak across train and holdout through different phases. By locking the split on groups, the benchmark preserves the structure of the generated motion sequence while still exposing the detector to a proper blind holdout. The balance tables also show that the benchmark is not a single-class toy: the negative families include birds, RC fixed-wing aircraft, weather cells, ground vehicles, wind turbines, multipath ghosts, RFI bursts, and clutter-only coun-

TABLE VIII
READER-FACING LOCKED COMPONENT WEIGHTS SORTED BY WEIGHT

Audit ID	Family	Modality	Calibrator	Weight	Cum.	View
C2	Passive quality	passive-RF provenance	geodesic odds	0.433	0.433	signed DE
C4	Transport geometry	motion/geometry	geodesic odds	0.292	0.725	positive DE
C8	Passive hypergraph	passive-RF graph	geodesic odds	0.153	0.879	signed DE
C1	Passive quality	passive-RF provenance	raw	0.037	0.916	signed DE
C5	Transport geometry	motion/geometry	raw	0.034	0.950	signed DE
C3/C6/C7	Transport/passive residual	mixed	mixed	0.050	1.000	residual

TABLE IX
COMPARABLE DETECTOR-VIEW CONTROLS ON THE BLIND HOLDOUT SPLIT

View/control	AP	ΔAP	Recall@≤1%FPR	ΔRecall	F1	ECE
full locked candidate	0.825	0.000	0.833	0.000	0.837	0.002
radar-only view	0.126	-0.699	0.250	-0.583	0.139	0.447
acoustic-only view	0.093	-0.732	0.208	-0.625	0.125	0.754
passive-RF-only view	0.847	+0.022	0.958	+0.125	0.732	0.228
radar+acoustic view	0.159	-0.666	0.333	-0.500	0.225	0.527
radar+passive-RF view	0.547	-0.278	0.625	-0.208	0.526	0.363

TABLE X
COMPARE-ONLY ANCHOR ROLES

Anchor	EchoForge use boundary
KTH 77 GHz FMCW	Compare-only hard-negative realism and observable-shape checks; no positive truth claim
Open Radar Initiative	Moving-object and micro-Doppler distribution checks after local access rules are satisfied
RAD-DAR/RDRD	Compact range-Doppler comparison after local terms are accepted
Han/Jung 2026	Future multimodal alignment checks, pending license review

terfactuals.

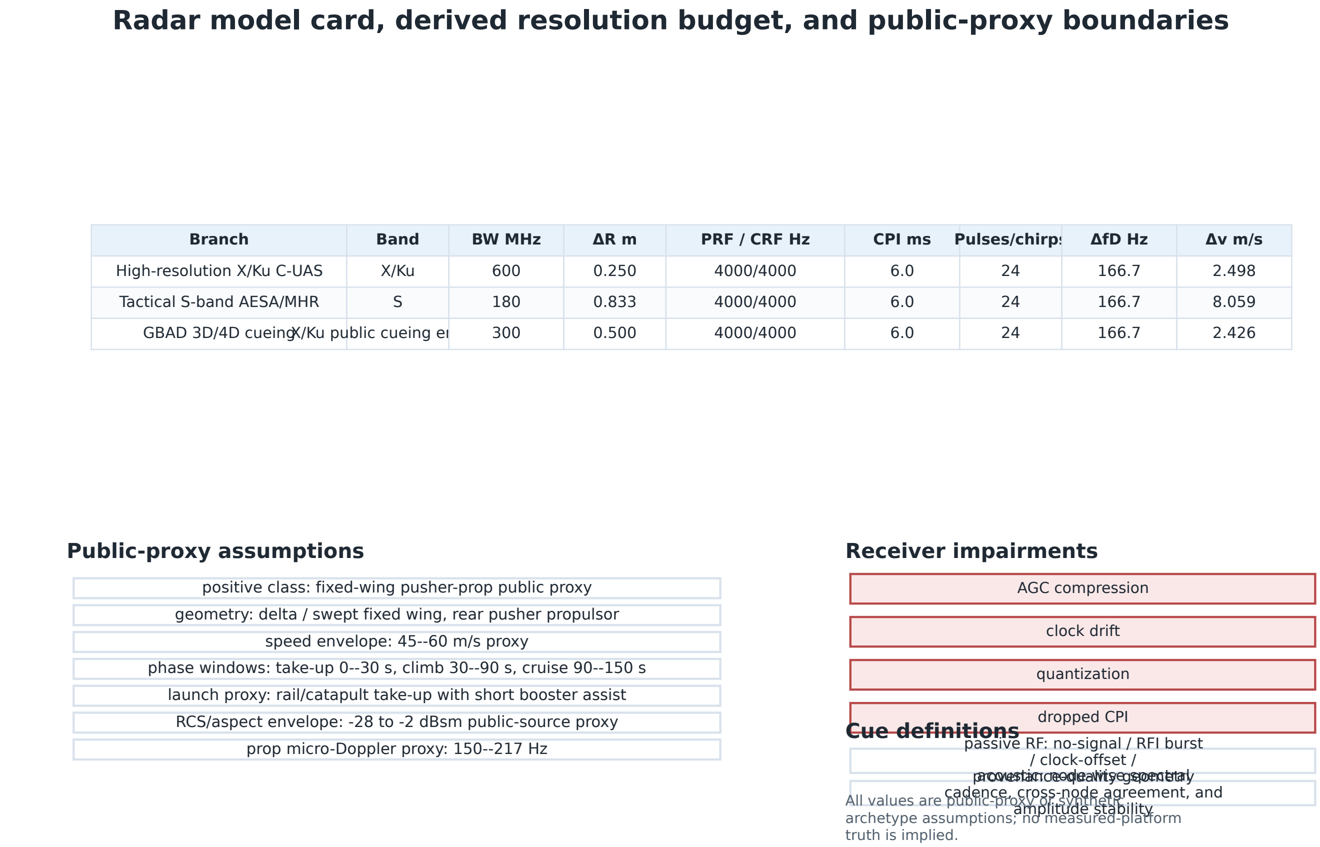
The evaluation bundle is equally explicit about how scores are read. Average precision and ROC AUC are reported alongside selected-threshold counts, fixed-FPR recall, group-level operating counts, calibration bins, Brier score, expected calibration error, and bootstrap intervals grouped by scenario. The group-block bootstrap is the right uncertainty model here because the group lock is part of the problem definition; a row bootstrap would overstate certainty by pretending that repeated phases from the same group are independent. This is why the paper reports both aggregate holdout performance and phase-stratified summaries.

The compare-only KTH overlay serves a narrower purpose. It is a measured anchor for observable-shape comparison, useful for checking whether the synthetic hard negatives and the public-proxy positive class live in a plausible neighborhood of micro-Doppler bandwidth, spectral entropy, and range-style observables. The evidence bundle records standardized feature deltas and Wasserstein-style gap rows where local anchor distributions permit them. The anchor does not make the synthetic fixed-wing class into a measured platform proxy, and it should not be read as such.

The locked-candidate transparency files complete the ledger by exposing the selected candidate as a locked composition. The selected component scores, aliases, human-readable weights, comparable view controls, and quarantined internal ablations make it possible to ask whether a paper figure is using the same metric basis as the main result. In the current run, the selected candidate's false alarms concentrate in a small number of hard-negative families rather than in an undifferentiated mass of failures. That is the kind of evidence a radar reviewer can inspect without being asked to trust a leaderboard jump on faith.

X. DETECTOR FAMILY APPENDIX

The evidence bundle includes appendix-ready JSON and CSV cards for each modeled detector/source family. The X/Ku C-UAS card records low-slow-small search, high-resolution range-Doppler, micro-Doppler, clutter suppression, and bird/RC discrimination assumptions. The tactical S-band AESA/MHR card records hemispheric surveillance, track-while-scan confidence, coarser velocity resolution, and multipath stress. The GBAD 3D/4D card records wide-area cueing, revisit delay, radar-horizon masking, and lower-resolution classification. The classical radar card records CA-CFAR, OS-CFAR, MTD concentration, M/N confirmation, and declared PFA. The acoustic card records spectral cadence, bearing/time correlation, node agreement, weather, traffic, and ambient-noise false cues. The passive-RF card records no-signal behavior, RFI bursts, emitter/provenance cues, and explicit missingness for RF-silent targets. The layered-fusion card records source confidence, stale-track handling, cross-sensor confirmation, and false-track accounting.



Branch	Band	BW MHz	ΔR m	PRF / CRF Hz	CPI ms	Pulses/chirps	ΔfD Hz	Δv m/s
High-resolution X/Ku C-UAS	X/Ku	600	0.250	4000/4000	6.0	24	166.7	2.498
Tactical S-band AESA/MHR	S	180	0.833	4000/4000	6.0	24	166.7	8.059
GBAD 3D/4D cueing	X/Ku public cueing en	300	0.500	4000/4000	6.0	24	166.7	2.426

Fig. 6. Radar model card and signal-chain parameter panel. The figure surfaces the public-proxy assumptions behind the benchmark so the detector results can be read against explicit carrier, bandwidth, PRF/CPI, and cue definitions.

XI. Limitations and Prohibited Inferences

The most important limitation is also the simplest: a synthetic benchmark is not measured truth. The present corpus should be read as public-proxy evidence with declared priors, not as a fielded sensor surrogate. The comparison against KTH and other public anchors is a compare-only check, useful for distribution shape, calibration sanity, and hard-negative realism, but insufficient to justify named-platform claims. That boundary follows the usual verification-and-validation distinction in computational science and the reproducibility recommendations for machine-learning research [32]–[35].

The statistical limitation is central. The corpus is a single-seed run with 50 positive groups total and only 8 positive groups, 24 positive records, in the holdout split. Group-block bootstrap intervals are therefore necessary but not sufficient for broad claims. Phase and false-alarm-family slices are robustness diagnostics, not stable estimates of field rates.

The physics limitation is also explicit. The radar branch cards use simplified public-proxy RCS, clutter, weather, RFI, waveform, and processing assumptions. KTH is a 77 GHz measured anchor with a different frequency and target mix, so it can support compare-only observable-shape checks but not measured positive-class validation. The paper contains no hardware-in-the-loop evidence, no measured positive-class validation, no track-level FAR, and no deployment-performance claim. The next useful steps are wider terrain and clutter priors, richer weather and RFI stressors, multi-seed corpora, larger positive holdouts, hardware-in-the-loop hooks, and finer-grained uncertainty reporting.

XII. Fixed-Wing Pusher-Prop Public-Proxy Appendix

The appendix-only public proxy card records the assumed geometry and flight envelope for the positive class. The boundary is intentionally narrow:

- public-source assumptions: delta or swept fixed wing, rear pusher propulsor, approximate 3.3–3.7 m length, 2.3–2.7 m wingspan, and 0.35–0.75 m height;
- public-source kinematics: 45–60 m/s speed envelope, take-up/climb/cruise phase windows, and a rail/catapult take-up proxy with short booster assist;
- public-source observables: an aspect-dependent RCS envelope and a prop micro-Doppler envelope that are only broad proxy bands, not measured signatures.

The appendix uses open visual reporting and public summaries [36]–[38] together with small-UAS radar literature [8], [9]. Prohibited inferences are equally explicit:

- no measured Iranian-drone radar signature;
- no operational route or evasion modelling;
- no payload-effects inference;
- no deployment-performance claim;
- no sensor-parity or classified-fidelity claim.

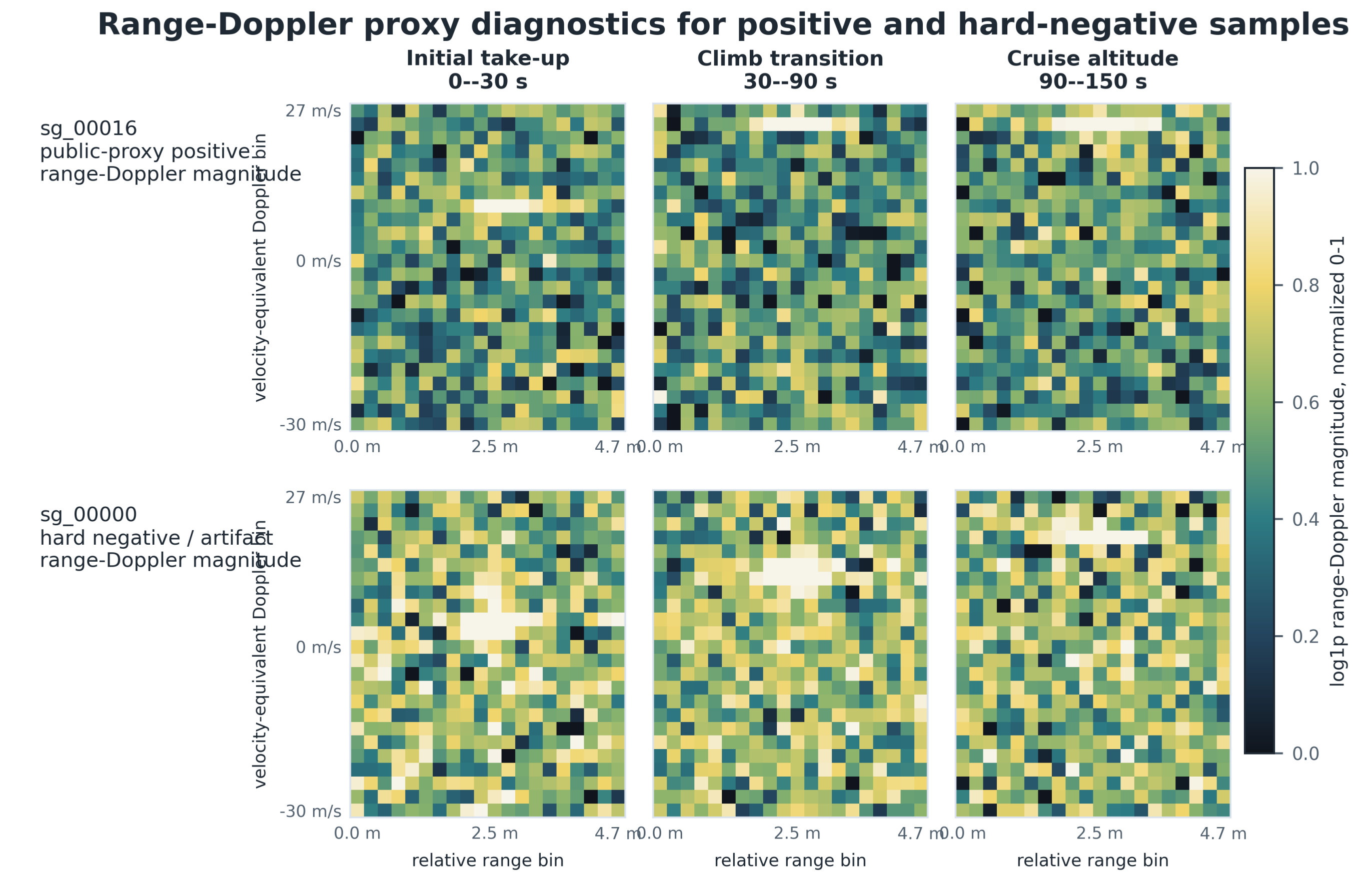


Fig. 7. Range-Doppler proxy diagnostics for a positive public-proxy group and a hard-negative group. Axes show relative range in meters and velocity-equivalent Doppler bins using the X/Ku branch resolution; color is normalized log-magnitude on a shared scale.

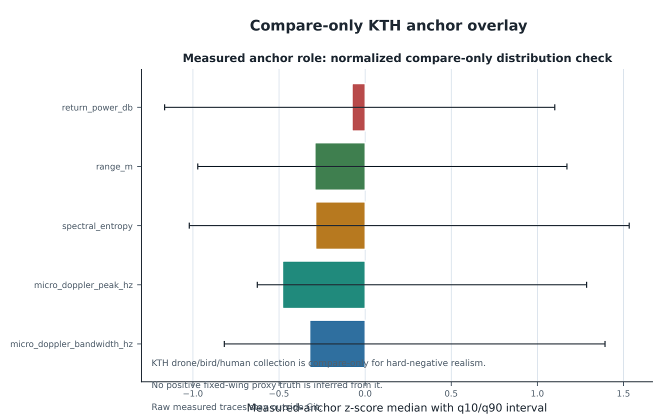


Fig. 8. Compare-only KTH anchor overlay. Features are normalized to measured-anchor z-scores before plotting, so the figure compares distribution shape rather than mixing raw physical units.

XIII. Regional Bird and RC Hard-Negative Appendix

The regional bird library is split into hard-negative families that are common around Gulf and coastal sites: single bird, bird flock, shorebird/wader, gull/tern, raptor/falcon, flamingo/large bird, seabird/cormorant, seasonal migratory density, and RC fixed-wing. The evidence bundle records wingbeat bands, body-size/RCS stress notes, site prevalence, seasonality, flock behavior, expected confusion mechanisms, and generated false-alarm summaries in `regional_bird_library.csv` and `source_pack_bird_coverage_summary.csv`.

The source-pack layer includes bird hard-negative cards for the single-bird, dense flock, shorebird/wader, gull/tern, raptor/falcon, flamingo/large-bird, seabird/cormorant, migratory-density, and RC fixed-wing entries, with the pack manifest updated to enumerate those cards explicitly. The compare-only bird literature is public and broad [8], [9], [19]; the point of the library is to expose likely false-positive pressure, not to optimize evasion.

XIV. Conclusion

EchoForge demonstrates a strict-open path for low-altitude UAS radar and multimodal sensing studies: public-proxy priors, deterministic generation, detector-view denylisting, group-locked holdout evaluation, group-block uncertainty estimates, compare-only measured-anchor checks, and tracked paper evidence. The headline result is LCB95 Recall@$\leq$1%FPR of 0.699 for the locked candidate, with point Recall@$\leq$1%FPR of 0.833 on the 24-positive-record, 8-positive-group holdout slice. The benchmark numbers in this paper are synthetic public-proxy evidence under declared assumptions. They are

TABLE XI
LEAKAGE AND REALISM DIAGNOSTICS

Diagnostic	Value	Interpretation
Raw audit-header detection + detector schema pass	Audit-only headers detected and blocked: `cv_fold`, `scenario_group_id`, `split_role`, and `time_lock_id`	The exported audit views expose these headers for guard testing; the detector-facing schema denylist blocks them before model scoring.
Label-shuffle sanity	Selected AP 0.825 vs shuffled AP 0.006; selected ROC AUC 0.917 vs shuffled ROC AUC 0.497; selected ECE 0.002 vs shuffled ECE 0.010	Real labels carry structure; shuffled labels collapse toward chance.
Stratum-only baseline	AP 0.005; ROC AUC 0.457; ECE 0.047	Split metadata alone is weak.
Metadata-only baseline	AP 0.004; ROC AUC 0.473; ECE 0.005	Coarse metadata alone is also weak.
Nearest-neighbor scan	Mean 0.1435, median 0.1226, p10 0.0703, p90 0.2419	Holdout rows are near, but not duplicates of, train rows in the detector-view space.

TABLE XII
SELECTED-THRESHOLD CONFUSION MATRIX FOR THE PRIOR FUSION BASELINE AND LOCKED CANDIDATE

Method	TP	FP	TN	FN	Precision	Recall	F1	FPR
layered_fusion_c2	10	49	4427	14	0.169	0.417	0.241	0.010947
locked candidate	18	1	4475	6	0.947	0.750	0.837	0.000223

TABLE XIII
CROSS-METHOD FALSE-POSITIVE BURDEN FOR THE STRONGEST FIVE METHODS

Method	AP	Recall@≤1%FPR	FP	FP/1000 Neg.	Top FP Family	Burden vs Locked
locked candidate	0.825	0.833	1	0.223	single_bird	0
layered_fusion_c2	0.128	0.292	49	10.947	rc_fixed_wing	48
tabular_ml_baseline	0.126	0.250	70	15.639	weather_cell	69
sequence_ml_proxy	0.116	0.250	7	1.564	weather_cell	6
high_resolution_xku_cuas	0.112	0.333	44	9.830	rc_fixed_wing	43

TABLE XIV
SELECTED CANDIDATE BY PHASE ON THE BLIND HOLDOUT SPLIT

Phase	N	Pos	AP	ROC AUC	Recall@≤1%FPR	TP	FP	TN	FN
Initial take-up	1500	8	0.628	0.813	0.625	4	0	1492	4
Climb transition	1500	8	0.845	0.937	0.875	6	1	1491	2
Cruise altitude	1500	8	1.000	1.000	1.000	8	0	1492	0

useful for review, robustness work, and reproducible comparison, but they are not measured truth or operational performance claims.

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TABLE XV
SELECTED CANDIDATE FALSE-ALARM FAMILIES

Family	Count	FA	Rate	Near
bird_flock	474	0	0.000	0
clutter_only	480	0	0.000	0
ground_vehicle	450	0	0.000	0
multipath_ghost	438	0	0.000	0
rc_fixed_wing	483	0	0.000	0
rfi_burst	387	0	0.000	0
single_bird	405	1	0.002	0
terrain_glint	414	0	0.000	0
weather_cell	486	0	0.000	0
wind_turbine	459	0	0.000	0

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